

BILKENT UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING

MATH 241
ENGINEERING MATHEMATICS

MIDTERM EXAM
JULY 9, 2014

Instructions:

- There are 5 questions in the exam.
- Duration of the exam is 120 minutes.
- The exam is closed book, closed notes.
- No calculators are allowed.
- Make sure that your cellphone is turned off before starting the exam.
- Please read each question carefully and explain your answers clearly in order to get credit.
- Do not detach pages.

14 16 17 18
21 23 24 25
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Surname : SOLUTION MANUAL
Name :
ID :
Signature :

Question	Grade
1 (20)	
2 (20)	
3 (20)	
4 (15)	
5 (25)	
Total: (100)	

Q1. (20 points) Consider the following system of linear equations.

$$\begin{aligned}x_2 + x_3 &= 5 \\x_1 + x_2 - x_3 &= 0 \\x_1 - x_2 + ax_3 &= b\end{aligned}$$

(a) Determine all possible values of the constants a and b so that the given system has

- (i) a unique solution.
- (ii) infinitely many solutions.
- (iii) no solution.

(b) Find the solution set in case that the system has infinitely many solutions.

Q1

$$\begin{bmatrix} 0 & 1 & 1 & 5 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & a & b \end{bmatrix} \xrightarrow[R_3 - R_1 \rightarrow R_3]{R_2 \leftrightarrow R_1} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 5 \\ 0 & -2 & a+1 & b \end{bmatrix} \xrightarrow{R_3 + 2R_2 \rightarrow R_3} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & a+3 & b+10 \end{bmatrix}$$

(a)

(i) $a \neq -3 \Rightarrow$ no free variable $\Rightarrow$ unique solution

(ii) $a = -3 \quad b = -10 \Rightarrow$ system is consistent

(iii) $a = -3 \quad b \neq -10 \Rightarrow$ system is inconsistent.

(b)

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - R_2 \rightarrow R_1} \begin{matrix} \begin{matrix} \downarrow & \downarrow & \downarrow \\ 1 & 0 & -2 & -5 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{matrix} \end{matrix} \Rightarrow x = \begin{bmatrix} -5 \\ 5 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} t$$

Q2. (20 points) Consider the following system of linear equations in the form of $Ax = b$,

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 1 & 2 & 3 & 1 \\ 1 & 3 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

- 5 (a) Using Gauss-Jordan elimination find the reduced row echelon form of the augmented matrix $[A \ b]$. Identify the pivot and free variables of the system.
 5 (b) Find the solution set of the system.
 5 (c) Find a basis for $Row(A)$, $Col(A)$ and $Null(A)$.
 5 (d) Find the particular solution that is orthogonal to $Null(A)$.

(a)
$$\begin{bmatrix} 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 3 & 1 & 2 \\ 1 & 3 & 4 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & 0 & 3 \\ 0 & 1 & 1 & 1 & -1 \\ 0 & 2 & 2 & 2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & -1 & 4 \\ 0 & 1 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$\begin{matrix} P & P & f & f \\ \downarrow & \downarrow & \downarrow & \downarrow \end{matrix}$
 x_1, x_2 pivots
 x_3, x_4 free variables

(b)
$$\underline{x} = \begin{bmatrix} 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

$x_3 = \alpha_1$
 $x_4 = \alpha_2$
 $x_1 = 4 - \alpha_1 + \alpha_2$
 $x_2 = -1 - \alpha_1 - \alpha_2$

(c)
$$Row(\underline{A}) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$
 non zero rows of the echelon form of $\underline{A}$.

$$Col(\underline{A}) = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$
 pivot columns of the original matrix $\underline{A}$

$$Null(\underline{A}) = \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$
 homogeneous solution.

(d) Need to find orthonormal basis for $Null(\underline{A})$

$$\hat{e}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{e}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad \underline{x} = \begin{bmatrix} 4 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$
 is a solution so find the orthogonal component of this particular solution.

Projected
$$\underline{x}_p = \sum_{i=1}^2 (\hat{e}_i \cdot \underline{x}) \hat{e}_i = -\frac{2}{3} \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + \frac{5}{3} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3+5 \\ 3-5 \\ -3+0 \\ 0+5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 8 \\ -2 \\ -3 \\ 5 \end{bmatrix}$$

orthogonal
$$\underline{x}_0 = \underline{x} - \underline{x}_p = \begin{bmatrix} 4 \\ -1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 8 \\ -2 \\ -3 \\ 5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 12-8 \\ -3+2 \\ 0+3 \\ 0-5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 \\ -1 \\ 3 \\ -5 \end{bmatrix}$$

Q3. (20 points)

- 10 (a) Let the orthonormal set of vectors $\{\hat{e}_1, \dots, \hat{e}_k\}$ spans the vector space S .
- 5 (i) Let $\underline{u}_p$ be the orthogonal projection of $\underline{u}$ onto S . Express $\underline{u}_p$ in terms of the set $\{\hat{e}_1, \dots, \hat{e}_k\}$ and $\underline{u}$.
- 5 (ii) Show that the error vector $\underline{\varepsilon} = \underline{u} - \underline{u}_p$ is orthogonal to S .

- 10 (b) Let $\underline{v}_1 = [2 \ 1 \ 2 \ 0]^T$, $\underline{v}_2 = [-2 \ 2 \ 1 \ 1]^T$, $\underline{v}_3 = [-1 \ 1 \ 5 \ 1]^T$, $\underline{v}_4 = [3 \ -3 \ 3 \ -1]^T$, and $V = \text{span}\{\underline{v}_1, \underline{v}_2, \underline{v}_3, \underline{v}_4\}$.
- 3 (i) Is the vector set $\{\underline{v}_1, \underline{v}_2, \underline{v}_3, \underline{v}_4\}$ a linearly independent set of vectors?
- 3 (ii) Find an orthonormal basis for V .
- 3 (iii) Let $\underline{u} = [1 \ -1 \ 4 \ 10]^T$, find $\underline{u}_p$ which is the orthogonal projection of $\underline{u}$ onto V .
- 2 (iv) Verify that the error vector $\underline{\varepsilon} = \underline{u} - \underline{u}_p$ is orthogonal to V .

(a) (i) $\underline{u}_p = \sum_{i=1}^k (\hat{e}_i \cdot \underline{u}) \hat{e}_i = \underline{E} \cdot \underline{E}^T \underline{u}$ $\underline{v} = \underline{E} \underline{\alpha}$ $\underline{E}$ $\underline{u}$

(ii) $\underline{\varepsilon} \cdot \underline{v} = (\underline{u} - \underline{u}_p) \cdot (\sum_{j=1}^k \alpha_j \hat{e}_j)$ $\underline{v} = \sum_{j=1}^k \alpha_j \hat{e}_j$ $\underline{v} \in S$

$$= \sum_j \alpha_j \hat{e}_j \cdot \underline{u} - (\sum_{i=1}^k (\hat{e}_i \cdot \underline{u}) \hat{e}_i) \cdot (\sum_{j=1}^k \alpha_j \hat{e}_j)$$

$$= \sum_j \alpha_j \hat{e}_j \cdot \underline{u} - \sum_i \sum_j \alpha_j (\hat{e}_i \cdot \underline{u}) \hat{e}_i \cdot \hat{e}_j$$

$\hat{e}_i \cdot \hat{e}_j = \delta_{ij}$

$$= \sum_j \alpha_j \hat{e}_j \cdot \underline{u} - \sum_j \alpha_j (\hat{e}_j \cdot \underline{u}) = 0$$

(b) (i) $\begin{bmatrix} 2 & -2 & -1 & 3 \\ 1 & 2 & 1 & -3 \\ 2 & 1 & 5 & 3 \\ 0 & 1 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & -1 & 3 \\ 0 & 3 & 3/2 & -9/2 \\ 0 & 3 & 6 & 0 \\ 0 & 1 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & -1 & 3 \\ 0 & 3 & 3/2 & -9/2 \\ 0 & 0 & 9/2 & 9/2 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 & -1 & 3 \\ 0 & 3 & 3/2 & -9/2 \\ 0 & 0 & 9/2 & 9/2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{LD}$ rank=3

(ii) $\hat{e}_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix}$, $\hat{e}_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} -2 \\ 2 \\ 1 \\ 1 \end{bmatrix}$, $\underline{v}_{3p} = \frac{1}{9} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \frac{1}{10} \begin{bmatrix} -2 \\ 2 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \\ 1 \\ 1 \end{bmatrix}$ # LI vectors = 3

$$\hat{e}_3 = \frac{1}{3} \begin{bmatrix} -1 \\ 2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 3 \\ 1 \end{bmatrix}$$

$$\underline{\varepsilon}_3 = \underline{\varepsilon}_3 = \underline{v}_3 - \underline{v}_{3p} = \begin{bmatrix} -1 \\ 1 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 3 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 2 \\ 0 \end{bmatrix}$$

(iii) $\underline{u}_p = \frac{1}{9} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \frac{1}{10} \begin{bmatrix} -2 \\ 2 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} -2 \\ 2 \\ 1 \\ 1 \end{bmatrix} + \frac{1}{9} \begin{bmatrix} -1 \\ 2 \\ 2 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 2 \\ 0 \end{bmatrix}$

$$= \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \\ 1 \\ 1 \end{bmatrix}$$

$\underline{D} = \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/10 & 0 \\ 0 & 0 & 1/9 \end{bmatrix}$

$\underline{v} = \alpha_1 \underline{e}_1 + \alpha_2 \underline{e}_2 + \alpha_3 \underline{e}_3$ $\underline{v} \in V$ $\underline{v} = \alpha_1 \underline{e}_1 + \alpha_2 \underline{e}_2 + \alpha_3 \underline{e}_3$ $\underline{v} = \begin{bmatrix} 2 & -2 & -1 \\ 1 & 2 & 2 \\ 2 & 1 & 2 \\ 0 & 1 & 0 \end{bmatrix}$

(iv) $\underline{\varepsilon} = \underline{u} - \underline{u}_p = \begin{bmatrix} 1 \\ -1 \\ 4 \\ 10 \end{bmatrix} - \begin{bmatrix} -1 \\ 5 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 3 \\ 9 \end{bmatrix}$

$$\begin{bmatrix} 2 \\ -2 \\ 3 \\ 9 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 & 2 & 0 \\ -2 & 2 & 1 & 1 \\ -1 & -2 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ -1 \\ 9 \end{bmatrix} = \alpha \begin{bmatrix} 1/3 & 0 & 0 \\ 0 & 1/10 & 0 \\ 0 & 0 & 1/9 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ -1 \\ 9 \end{bmatrix} = \alpha \begin{bmatrix} 2/3 & -2/10 & -1/9 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \alpha \underline{0} = 0$$

Q4. (15 points)

Let A be an n by n square matrix and $A = [a_1 \dots a_j \dots a_n]$ where a_j is the j^{th} column of A ($1 \leq i, j \leq n$).

- (3) (a) Using the cofactor expansion show that multiplying a column of A with a constant α results in multiplying the $\det(A)$ with the same constant α ,
i.e. $\det([a_1 \dots \alpha a_j \dots a_n]) = \alpha \det(A)$.
- (3) (b) Show that if one column of A is identical to another column of A then $\det(A) = 0$,
i.e. if $a_r = a_j$ then $\det(A) = 0$, $1 \leq r \leq n$ and $r \neq j$.
(Hint: Interchanging of any two columns of A changes the sign of the determinant.)
- (3) (c) Using the cofactor expansion show that,
 $\det([a_1 \dots a_{j-1} a_j a_{j+1} \dots a_n]) = \det([a_1 \dots a_{j-1} x a_{j+1} \dots a_n]) + \det([a_1 \dots a_{j-1} y a_{j+1} \dots a_n])$ where $a_j = x + y$.
- (8) (d) Using the previous parts show that if one column of A is a linear combination of other columns then $\det(A) = 0$, i.e. if $a_j = \sum_{r \neq j} \alpha_r a_r$ then $\det(A) = 0$.

Bonus (10 Points)

- (e) Using the cofactor expansion show that interchanging of two adjacent columns results in changing the sign or polarity of the determinant.
 $\det([a_1 \dots a_{j+1} a_j \dots a_n]) = -\det([a_1 \dots a_j a_{j+1} \dots a_n])$.
- (f) Interchanging of any two columns can be expressed as a sequence of interchanging of two adjacent columns. Using part (e) show that interchanging of any two columns changes the sign of the determinant.
 $\det([a_1 \dots a_j \dots a_r \dots a_n]) = -\det([a_1 \dots a_r \dots a_j \dots a_n])$.

$$(a) \det([a_1 \dots \alpha a_j \dots a_n]) = \sum_k \alpha a_{kj} A_{kj} = \alpha \sum_k a_{kj} A_{kj} = \alpha \det(A) \quad \text{where } a_r = a_j, a_j = a_j$$

$$(b) \det([a_1 \dots a_r \dots a_j \dots a_n]) = -\det([a_1 \dots a_j \dots a_r \dots a_n]) = -\det([a_1 \dots a_r \dots a_j \dots a_n]) - \det(A)$$

$$\det(A) = -\det(A) \Rightarrow 2\det(A) = 0 \Rightarrow \det(A) = 0$$

$$(c) \det([a_1 \dots x+y \dots a_n]) = \sum_k (x_k + y_k) A_{kj} = \sum_k x_k A_{kj} + \sum_k y_k A_{kj} = \det([a_1 \dots x \dots a_n]) + \det([a_1 \dots y \dots a_n])$$

$$(d) \det([a_1 \dots \sum_{r \neq j} \alpha_r a_r \dots a_n]) \stackrel{(c)}{=} \sum \det([a_1 \dots \alpha_r a_r \dots a_n]) \stackrel{(a)}{=} \sum \alpha_r \det([a_1 \dots a_r \dots a_n]) \stackrel{(b)}{=} \sum \alpha_r \cdot 0 = 0$$

$$(e) \sum \alpha_{kj} \det([a_1 \dots a_{j+1} a_j \dots a_n]) = \sum \alpha_{kj} B_{k(j+1)} = \sum \alpha_{kj} - A_{kj} = -\sum \alpha_{kj} A_{kj} = -\det(A)$$

$$B_{k(j+1)} = (-1)^{k+j+1} N_{k(j+1)} = (-1)^{k+j+1} M_{kj} = -A_{kj}$$

minor of new matrix

minor of old matrix

for $j, j+1$ these submatrices does not change so does minors!

(b) let r th column be $\underline{x}$
 j th column be $\underline{y}$

position

$\underline{x}$ $\underline{y}$

moving $\underline{x}$ to $j-1$ st column $(j-1-r)$ swaps of adjacent columns $j-1$ j

moving $\underline{y}$ to $j-1$ st column 1 swap (Actually with $\underline{x}$) j $j-1$

moving $\underline{y}$ to r th column $(j-1-r)$ swaps of adjacent columns j r

so
$$\det([a_1 \dots a_r \ a_j \dots a_n]) = \underbrace{(-1)^{2(j-1-r)+1}}_{-1} \det([a_1 \dots a_j \dots a_r \dots a_n]) \quad \blacksquare$$

Q5. (25 Points)

- 6 (a) Prove that the real part of the eigenvalues of a real skew symmetric matrix A is zero.
 5 (b) Let columns of the square matrix, Q , be orthonormal. Show that the rows of Q are also orthonormal.

(c) Consider the matrix A .

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

7 (i) Compute $A^4 - 3A^3 - A^2 + 4A - 2I$.

7 (ii) Compute $\sin(\frac{\pi}{2}A)$.

$$\begin{aligned} (a) \quad x^H A x &= \lambda x^H x \\ &= (-A^T x)^H x = -\lambda^* x^H x \end{aligned} \Rightarrow \underbrace{(\lambda + \lambda^*)}_{=0} \underbrace{x^H x}_{\neq 0} = 0$$

$$\lambda = -\lambda^* \Rightarrow \operatorname{Re}\{\lambda\} = 0$$

$$(b) \quad Q^T Q = \underline{I} \quad \text{since} \quad Q = [e_1 \dots e_n] \quad e_i^T e_j = \delta_{ij}$$

Q has LI columns so Q^{-1} is unique $Q^{-1}Q = \underline{I}$

Since Q^{-1} is unique $Q^{-1} = Q^T$

$$Q Q^{-1} = Q Q^T = \underline{I} \Rightarrow \begin{bmatrix} r_1^T \\ r_2^T \\ \vdots \\ r_n^T \end{bmatrix} [c_1 \ c_2 \ \dots \ c_n] = \underline{I} \Rightarrow r_i^T c_j = \delta_{ij}$$

$\Rightarrow r_i$'s are orthonormal

$$(c) \quad \det(A - \lambda I) = (1-\lambda)((1-\lambda)^2 - 4) = (1-\lambda)(\lambda^2 - 2\lambda - 3) = (1-\lambda)(\lambda+1)(\lambda-3)$$

$$= -\lambda^3 + 3\lambda^2 + \lambda - 3$$

$$\lambda_1 = 1 \quad \lambda_2 = -1 \quad \lambda_3 = 3$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - \lambda + 3 = 0$$

$$A^3 - 3A^2 - A + 3I = 0$$

$$\Rightarrow A^4 - 3A^3 - A^2 + 4A - 2I$$

$$= A(A^3 - 3A^2 - A + 3I) + A - 2I = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 4 & 1 \end{bmatrix} - \begin{bmatrix} 2 & & \\ & 2 & \\ & & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 1 & 2 \\ 0 & -1 & 1 \\ 0 & 4 & -1 \end{bmatrix}$$

$$\sin\left(\frac{\lambda}{2}A\right) = \beta_2 \underline{A^2} + \beta_1 \underline{A} + \beta_0$$

$$\sin\left(\frac{\pi}{2}\lambda\right) = \beta_2 \lambda^2 + \beta_1 \lambda + \beta_0$$

$$\left. \begin{aligned} \sin\left(\frac{\pi}{2}\right) = 1 &= \beta_2 + \beta_1 + \beta_0 \\ \sin\left(-\frac{\pi}{2}\right) = -1 &= \beta_2 - \beta_1 + \beta_0 \end{aligned} \right\} \begin{aligned} \beta_1 &= 1 \\ \beta_0 &= -\beta_2 \end{aligned}$$

$$\sin\left(\frac{3\pi}{2}\right) = -1 = 9\beta_2 + 3\beta_1 + \beta_0$$

6

$$9\beta_2 + 3 - \beta_2 = -1 \quad 8\beta_2 = -4 \quad \beta_2 = -\frac{1}{2}$$

$$\beta_0 = \frac{1}{2}$$

$$A^2 = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 8 & 5 \end{bmatrix}$$

$$\begin{aligned} \sin\left(\frac{\lambda}{2}A\right) &= -\frac{1}{2} \begin{bmatrix} 1 & 10 & 5 \\ 0 & 5 & 2 \\ 0 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 4 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -4 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 0 \\ 0 & 0 & -1 \end{bmatrix} \end{aligned}$$

$$\sin(x) = \sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{n-1} \beta_n x^n \quad Q = [e_1, e_2, e_3] \rightarrow \text{eigenvalues}$$

$$x^n = \sum_{n=1}^{\infty} \beta_n x^n$$

n is size of the matrix.

$$Q^{-1} f(\lambda) Q$$

$$\lambda^n$$

$$p(\lambda) \rightarrow \text{root } \lambda_1, \lambda_n \quad p(\lambda_i) = 0$$

$$Q^{-1} \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} Q$$

$$\frac{q(\lambda)}{r(\lambda)} \mid \frac{p(\lambda)}{d(\lambda)}$$

$$\begin{array}{r} x^2 + x \mid x^2 + 2 \\ x^2 + 2x \mid x \neq 1 \\ \hline -x \\ \hline -x - 2 \end{array}$$

$$q(\lambda) = p(\lambda) d(\lambda) + r(\lambda)$$

$$q(\lambda_i) = p(\lambda_i) d(\lambda_i) + r(\lambda_i) \quad x^2 + x = (x+2)(x-1) + 2$$